CAMUL Project2

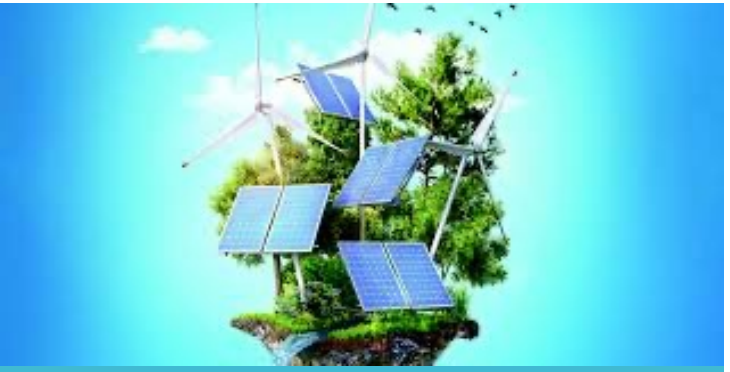
Submitting a paper

GAMES INTERACTION AND LEARNING TECHNOLOGIES

isep Instituto Superior de Engenharia do Porto

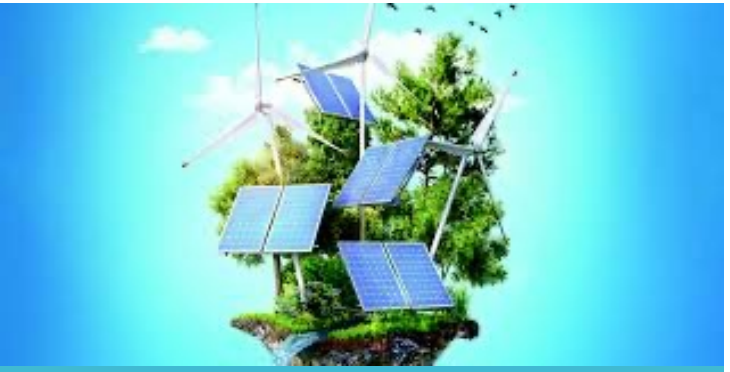
P2 Purpose:

- Consider submitting a paper to the CAMUL.
- Context: 1st Project development
- This work is directly related to Project 1: (example) the development of a MOOC on renewable and sustainable energy.
- Now, students must **reflect** on this practical experience and explore a related topic, with a focus on **teaching** or **technology**, in a scientific paper.



P2 Description:

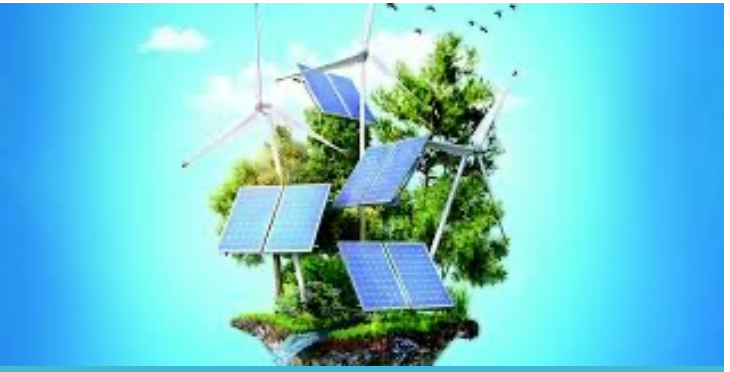
- The papers should address subjects such as:
  - Technologies used or proposed in the project, such as Matterport or multimedia tools, and their application in education.
  - Pedagogical strategies developed or suggested for the MOOC.
  - Innovations or challenges identified in the creation and integration of the course in MOOC platforms.
- Language: English
- Template: The required template is available on the IEEE website. The Microsoft Word A4 template (.docx format) is recommended.
- Length: The maximum length of the complete paper will be 6 pages with a minimum of 4 pages.



P2 Requirements:

- Team: 4 students per team (=P1).
- Focus: each team must choose a clear and coherent focus for the paper, selecting one of the two thematic directions:
  - **Teaching**-focused paper: e.g., pedagogical strategies, learning design, student engagement, instructional approaches in MOOCs
  - **Technology**-focused paper: e.g., Matterport, multimedia tools, VR/AR, digital platforms, technical integration challenges

  The chosen focus must remain consistent throughout the entire paper.



P2 Requirements:

- Individual Contribution: each team member must complete a significant and identifiable part of the work.

  Examples may include:

  - drafting specific sections of the paper
  - conducting part of the literature review
  - analysing data
  - contributing to the development of multimedia elements
  - preparing figures, tables, or diagrams

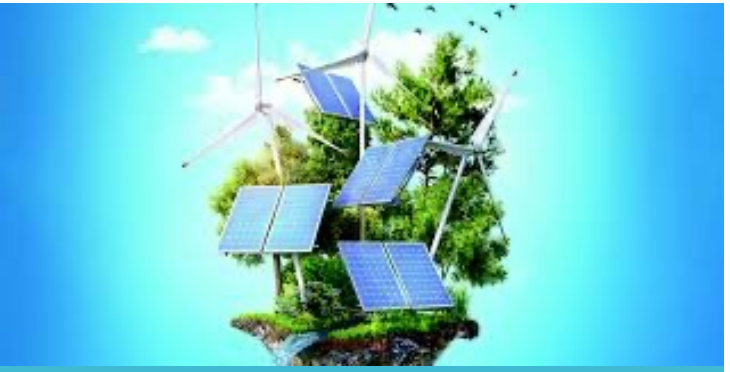
  Contributions should be balanced and fairly distributed among all members.

  Each individual contribution must be integrated into a unified and coherent final paper.

  The team is responsible for:

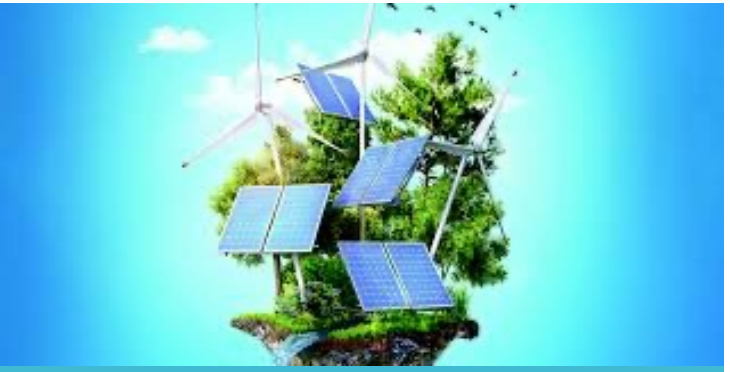
  - editing for consistency in tone, format, and terminology
  - ensuring smooth transitions between sections
  - validating that all contributions align with the chosen focus
  - checking compliance with the IEEE template and formatting guidelines

  The final document must read as one cohesive academic paper, not a collection of separate parts.



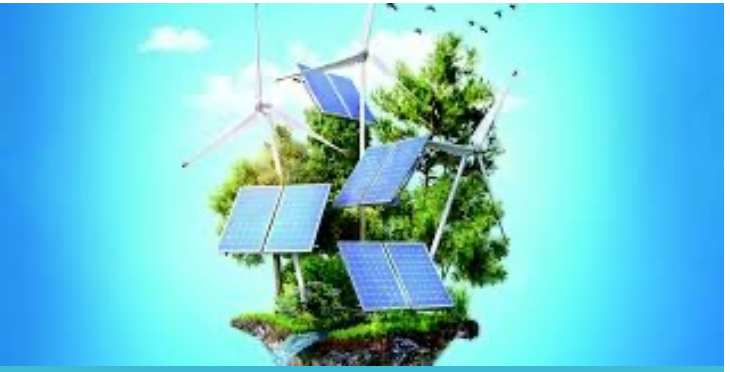
P2 Requirements:

- Collective Responsibility: although tasks are divided, the entire team is responsible for:
  - the quality and accuracy of the full paper
  - the correctness of references and citations
  - adherence to the 4–6 page limit
  - preparing and delivering the 10-minute presentation
- Originality and Academic Standards
  - Plagiarism is strictly prohibited.
  - All sources must be properly cited (minimum 13 scholarly references, last 5 years).
  - The final paper must follow academic writing conventions in English.



P2 Final Delivery :

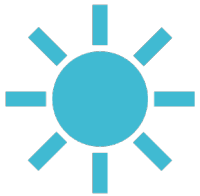
- Each team must submit a zip to the UC's Moodle with the materials developed, which include:
- The final work must be submitted as a .zip file, containing:
  - - The paper in Word and PDF formats.
  - - All graphs, tables, figures, photos and diagrams in separate files, with good resolution, numbered in the order in which they appear in the paper.
  - - The presentation in .ppt format.
- Face-to-face presentation during the PL class (mandatory) (cf., planning in Moodle).



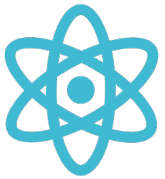
P2 Rating:

Category	Criteria	Weight	Performance Levels
1. Structure & Organization	IEEE format, 4–6 pages, mandatory sections, coherence	**15%**	Excelente (15), Bom (12), Suficiente (9), Insuficiente (6), Fraco (3)
2. Scientific Content & Depth	Relevance, accuracy, depth, alignment with chosen focus (teaching/technology)	**25%**	Excelente (25), Bom (20), Suficiente (15), Insuficiente (10), Fraco (5)
3. Related Research (Literature Review)	≥13 references (≤5 years), scientific quality (IEEE, ACM, Springer, Elsevier), integration	**15%**	Excelente (15), Bom (12), Suficiente (9), Insuficiente (6), Fraco (3)
4. Methodology & Data Analysis	Research model, procedures, data collection, qualitative/quantitative analysis	**15%**	Excelente (15), Bom (12), Suficiente (9), Insuficiente (6), Fraco (3)
5. Results & Discussion	Clarity, relevance, interpretation, comparison with literature, visuals	**15%**	Excelente (15), Bom (12), Suficiente (9), Insuficiente (6), Fraco (3)
6. Teamwork & Integration	Balance of contributions, coherence, collaboration, unified writing	**10%**	Excelente (10), Bom (8), Suficiente (6), Insuficiente (4), Fraco (2)
7. Presentation (10 minutes)	Structure, design, communication, time management, all members participate	**5%**	Excelente (5), Bom (4), Suficiente (3), Insuficiente (2), Fraco (1)
TOTAL		**100%**	

P2 – Paper Guidelines



- MOOC on Renewable Energy



- Scientific paper connected to Project 1



- Focus: pedagogy, technology, innovation

1. Introduction

- Contextualize MOOC and renewable energy

- Challenges from Project 1

- Technologies used (Matterport, multimedia)

Should include:

- •Context of the MOOC on Renewable Energy
- •Importance of digital learning and MOOCs today
- •Challenges and opportunities in integrating multimedia/technology
- •Clear organization of the paper (optional paragraph)

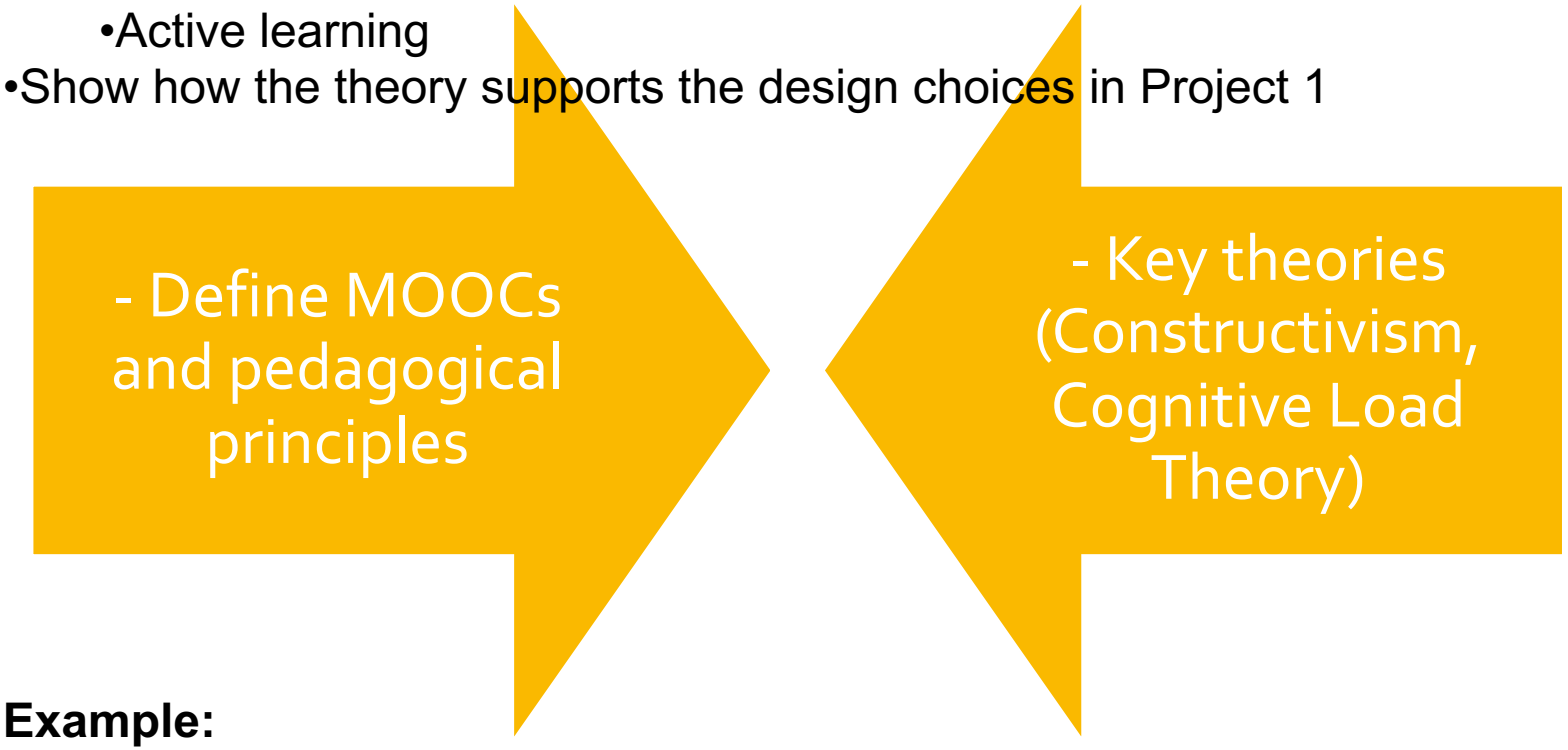
Example points:

•“Renewable energy education through digital platforms helps democratise access.”

•“The project implemented immersive tools such as Matterport to enhance engagement.”

Notes: Define the educational problem and justify why it is important.

•Define key concepts: MOOCs, digital learning, renewable energy education
•Relevant theories:
    •Constructivism
    •Multimedia learning (Mayer)
    •Active learning
•Show how the theory supports the design choices in Project 1

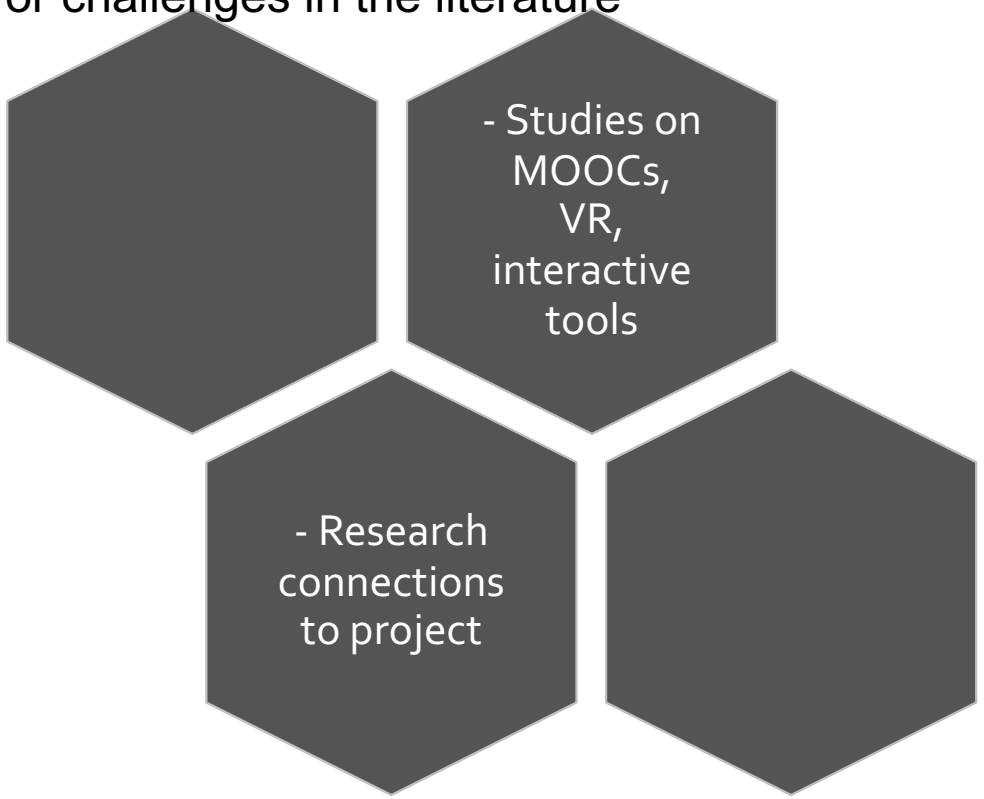


Example:
“According to Mayer’s multimedia principles, combining visuals and narration enhances comprehension.”

1.1.
Conceptual/Theoretical Framework

1.2. Related Research

- Review recent (≤5 years) studies on:
  - MOOCs and e-learning
  - Renewable energy education
  - Tools used (e.g., Matterport, 360º tours, multimedia)
- Identify gaps or challenges in the literature



Example:
“Recent studies highlight the need for interactive elements to increase learner motivation in online courses.”

1.3. Study Objective

- State a clear objective

- Include 2–3 research questions

Clearly define:

- What the paper aims to analyse
- Connection to Project 1

- Examples:
  - Evaluate the use of Matterport in a renewable energy MOOC
  - Analyse pedagogical strategies used in the course design
  - Identify challenges faced and propose improvements

Notes: Objectives should be simple, measurable, and aligned with the content that can be effectively analyzed.

2. Method and Materials

- Describe methodology and general approach



Should describe *how* the study was done:

- •Research model (qualitative, exploratory, design-based…)
- •Tools and platforms used (Matterport, video tools, LMS, etc.)
- •Steps in the MOOC creation process

Notes: Must demonstrate "methodological awareness," even if the project is small-scale.

2.1. Research Model

- Explain model: case study, DBR, exploratory



Possible models:

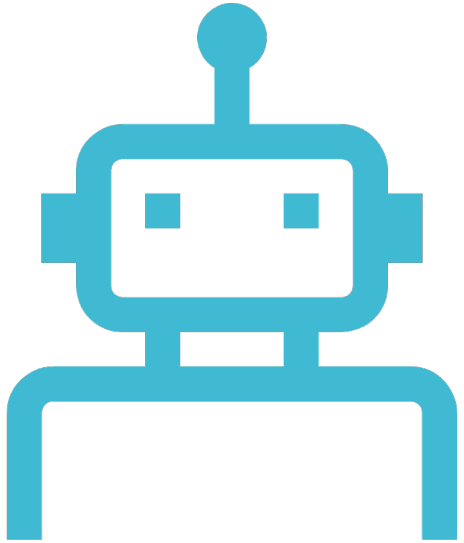
- •Exploratory study
- •Design-based research
- •Case study (the MOOC)
- •Usability/engagement analysis

Example:

“This work follows an exploratory design-based research model, focusing on the iterative creation of a MOOC unit.”

2.2. Methodology

- Participants, tasks, phases of development



Include:

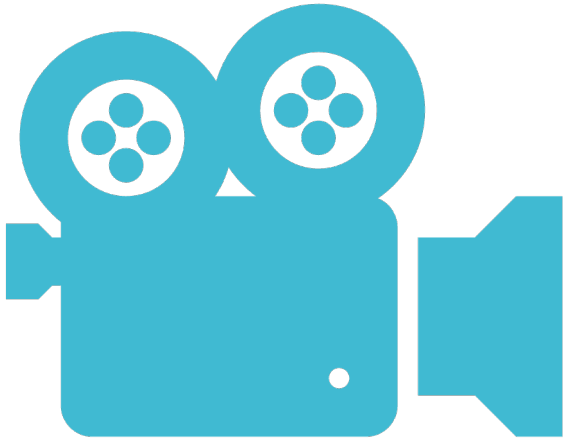
- •Description of tasks each team performed
- •Tools used for data gathering (*logs, feedback, peer review*)
- •Rationale for methodological choices

Example:

“Students applied multimedia learning principles to design instructional videos and interactive content.”

2.3. Development

- Technologies: Matterport, multimedia tools
- Process of building MOOC content



•Describe the development of:
    - •Scripts
    - •Multimedia content
    - •Matterport environment or virtual tours
    - •LMS integration
- •Screenshots recommended in appendix or as figures

Example:

“The team created a 3D virtual tour explaining solar panel components using Matterport.”

2.4. Data Collection

- Questionnaires, analytics, feedback



Examples of data sources:

- Observational notes from group work
- Feedback from peers or test users
- Analytics from platform (if any)
- Self-reflection logs

Notes:

If no “real data” exists, self-reported or design data is acceptable.

2.5. Data Analysis

- Quantitative + qualitative approaches



Should explain:

- •How data was interpreted
- •What categories or criteria were used

•Examples:

- Engagement indicators
- Usability issues
- Effectiveness of multimedia materials

3. Results



- MAIN FINDINGS



- GRAPHS OR TABLES WHEN APPROPRIATE

Show the main findings:

- •What worked well
- •What problems emerged
- •Evidence from the analysis

Example:

“Matterport improved visual comprehension but increased development time.”

Notes: Use of tables, figures, and concrete examples.



4. Discussion

- Interpretation of findings

- Comparison with related research

This section should:

- Compare results with the literature
- Explain WHY results occurred
- Identify key lessons learned

Example:

“Consistent with Mayer’s principles, the integration of 3D models enhanced understanding by providing spatial context.”

5. Conclusion

- Main lessons learned

- Limitations of the study

Include:

- •Summary of contributions
- •What the team learned from Project 1 and Project 2
- •Educational or technological relevance

Example:

“The study demonstrates the potential of immersive multimedia to strengthen MOOC effectiveness.”

6. Future Work

- IMPROVEMENTS FOR THE MOOC

- EXTENSIONS AND NEW MODULES

Ideas for improvement:

- Integration of quizzes inside Matterport
- Testing with real learners
- Adding adaptive learning
- Improving video quality or interactivity

Notes: Future work must be *realistic* and *feasible*.

- 13–20 references

- From 2020 onwards

- Use reputable sources

References

(IEEE style)